Image analysis (MicroICS) hands-on session: exercises

Part of: "Deep dive into the statistical approaches used to validate potential standardised laboratory methods"

In this session we will use the tool MicroICS to analyse 3D images of biofilms. This tool consists of three interoperable modules: 1) Feature calculation – extracting structural features from 3D confocal fluorescence images; 2) Model training – training machine learning models on those features with respect to your biological question; and 3) Inference – applying the trained model to unknown images to predict the property of the strain in the image.

The golden thread running through the image analysis tool MicroICS follows the pattern below, and it applies to your own biological question just as much as to questions studied on the Training School (TS):

Biological question → planning and image acquisition → linking image and biological question through labels → calculation of structural features → model training → answer to what separates the groups → prediction on unknown (treated) images → answer to whether a treatment changes the structure.

The biological question for this TS hands-on will be: can we distinguish between the eight *Listeria* strains solely on the basis of their 3D structure, and what causes the differences? Most of the steps carried out will be used to demonstrate how to use the software, and the results are illustrative; to obtain reliable answers to this question you will also need to complete the homework, where you will analyse the whole dataset.

The main takeaway from this session is that you can formulate a biological question that fits your area of interest and plan an analytical approach using the toolset presented.

The sample images, trained models, and results are available at https://portal.ijs.si/nextcloud/s/ZK847GAGxgWyjHo; please download them before starting the analysis.

Objective 1: Turn a biological question into a label

The label is the ground truth that you attach to each image, written into its filename (image) after _st_. Change the label and you change the question; the pipeline never changes.

Do: open the training image folder and look at the filenames. Find the _st_ part. That is the label.

Think and write: How would you name the images to distinguish clinically relevant from non-relevant strains? How would you label images for your question?

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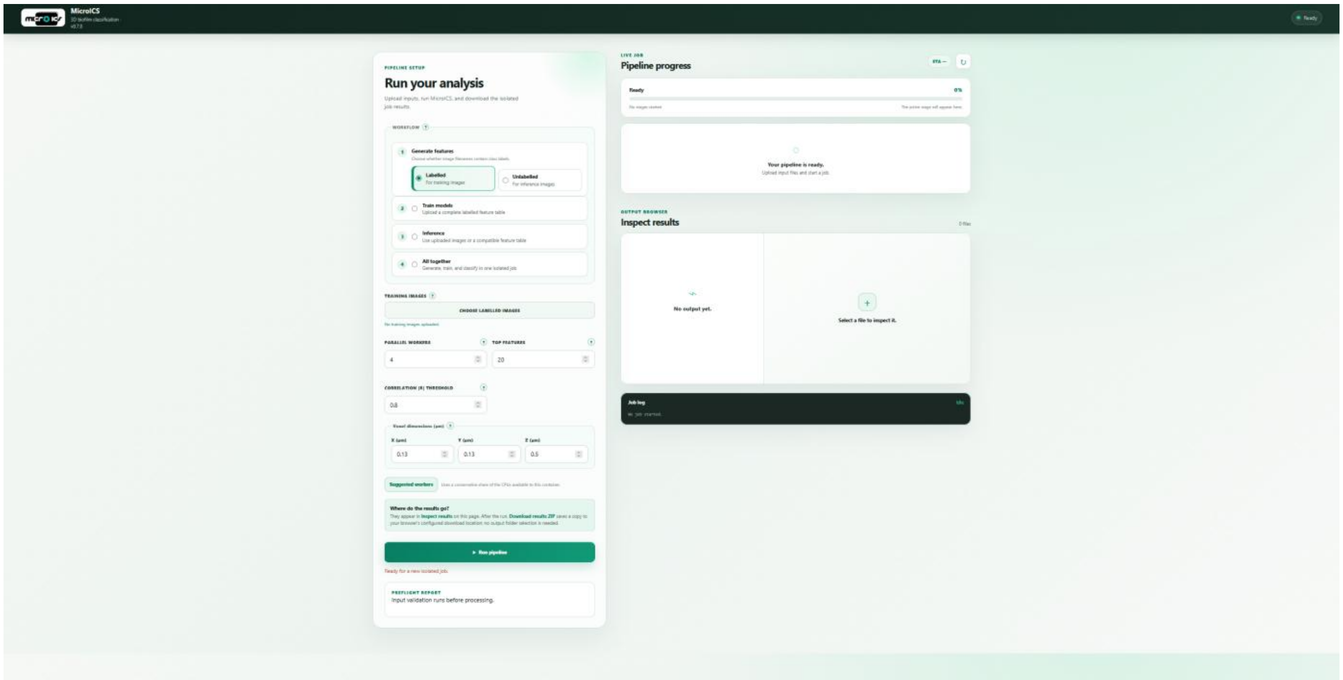
Take home: your question becomes a machine learning task the moment you can write its answer into the filename.

Objectives 3 – 5: Install and run MicroICS independently

Installation

- Open the document “Installation Guide” and start MicroICS (**Restarting the existing container of MicroICS**). The window that opens should look like this:

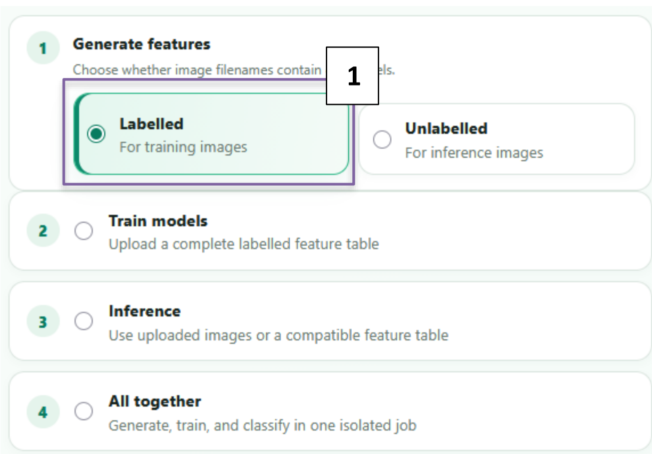


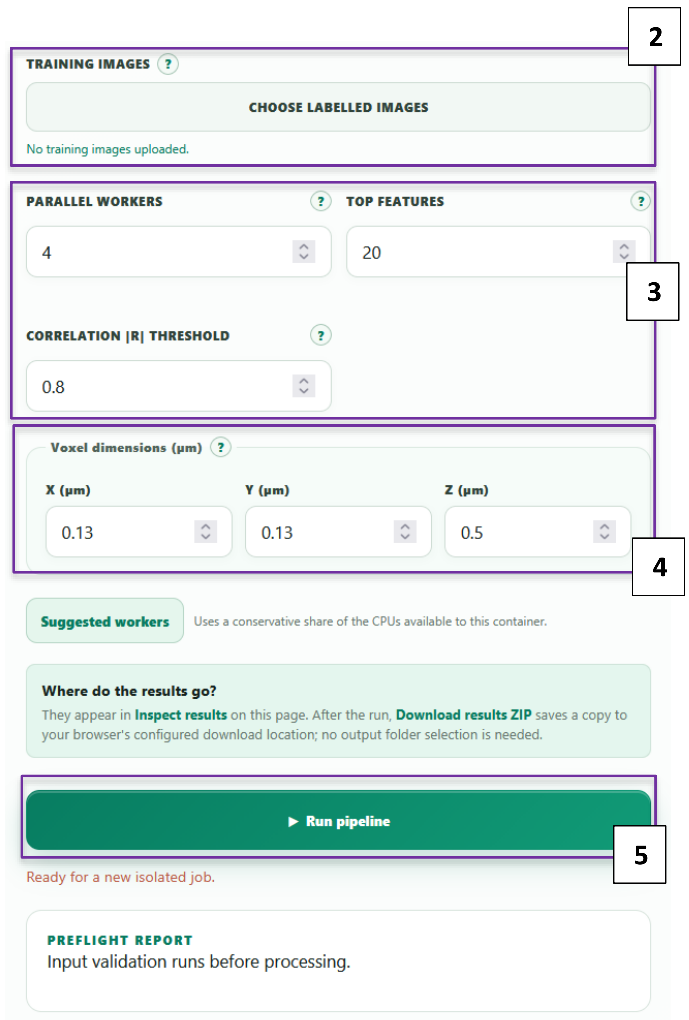
Generate features

To run “generate features”, you will need sample images from the folder images_generate_features_labelled.

Do:

1. Choose “Generate features” in mode “Labelled”. Labelled mode extracts the label from each filename, so that the model can link the features to the label during training.
2. Point the software to the folder containing your images in TIFF format, using “Choose labelled images”.
1. Set the running options. “Parallel workers” – how much of your CPU to dedicate to the analysis: use full capacity when you are not otherwise using the computer, and less if you are working on it at the same time. Suggested workers will provide optimal settings for running. At this point, “Top features” and “Correlation |R| threshold” do not matter.
3. Adjust the voxel dimensions (the pixel size in micrometres) to match your images.
4. Click “Run pipeline”. When the run starts, the software also checks the image naming and reports the number of classes it found (“Preflight report”).





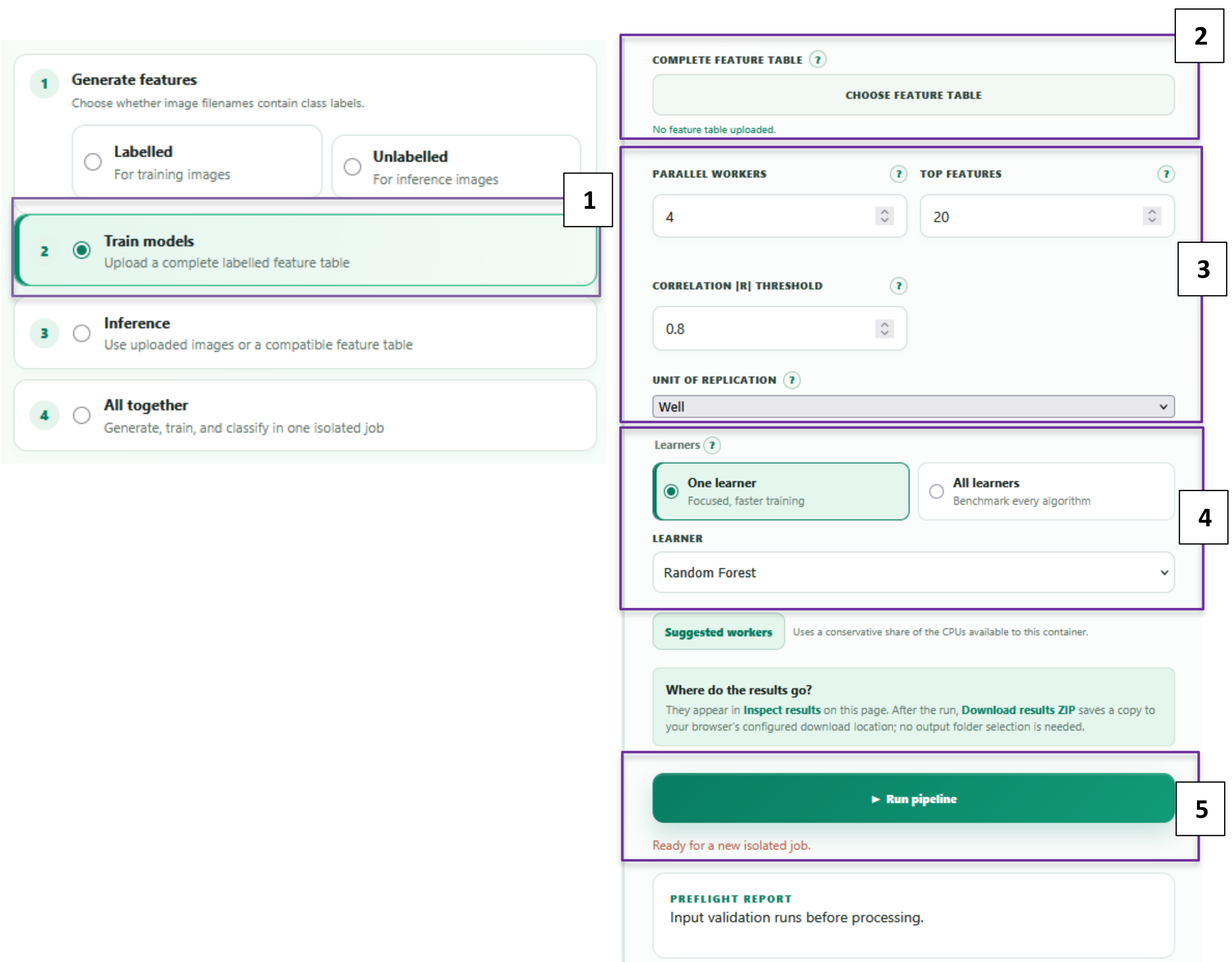
See the results: View the results by downloading the ZIP file. Open datafile.tsv – it has one row per image and roughly 2,700 columns, each representing a structural measurement (shape, texture, intensity, or spatial arrangement).

Train a model

Training the model to output meaningful results takes too long, so we provide a pre-made model for the training school and do not spend the session waiting. **Skip this part for now, but not for the homework. The pre-made models are available in the folder trained_models_identity.**

Do:

2. Choose "Train models."
3. Navigate to the folder containing calculated features (datafile.tsv) using "Choose feature table".
4. Set the running options: "Parallel workers" – how much of your CPU to dedicate to the analysis: full capacity when you are not otherwise using the computer, and less if you are working on it at the same time. Suggested workers will provide optimal settings for running. "Correlation threshold" – choose how many of the most important features to analyse, and the threshold for the analysis.
   "Unit of replication" – specify what counts as an independent experiment (day, plate, well, or position within the well). This is important for testing the trained model correctly.
5. Choose which learner to train on your data: a single learner and choose learner (model) from dropdown menu, or all those currently available.
6. Click "Run pipeline." When the run starts, the software also checks the input datafile for missing values (NaN, empty cells, etc.).



For now, focus on these outputs and set the rest aside:

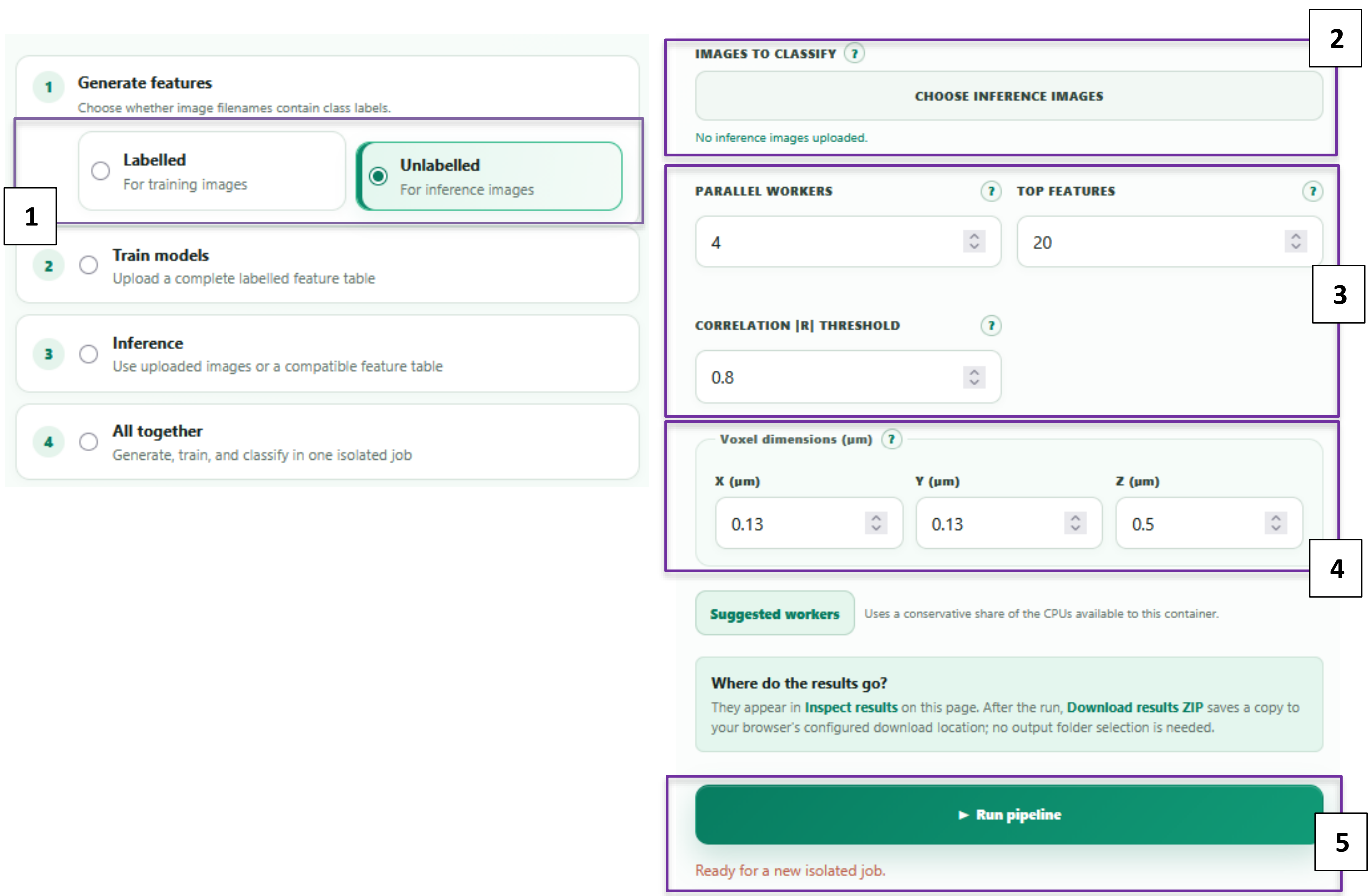
- **confusion matrix -** A matrix of true vs predicted class for the given dataset. Rows represent the true class, columns the predicted class; the diagonal shows correct predictions, and off-diagonal cells show errors. From this matrix you can identify which classes are easy to separate and which are confused with each other, implying that with the given set of features and model they are hard to distinguish.
- **rankings_label** – The structural features that best distinguish the strains. Here you can find information on which structural properties differ between the strains.
- **rankings_date** – Compared with rankings_label, it identifies features that also predict the imaging date; any feature ranking highly in both is tracking acquisition as much as biology and should therefore be interpreted with caution.
- **top_feature_correlations -** Correlation among the top features used to assess redundancy. Features are clustered according to their correlations, based on their values and the given cutoff. Features that fall within the same cluster may carry the same feature signal and thus be redundant. Examine the clusters and identify which features occur together. If the feature family imply the same trends, the clusters can be used for interpretation rather than individual features.

Inference

Inference consists of two steps: first, calculate features from the new images to obtain a list of features that the model can then use to make an inference on them, which is done in the second step. **To run "generate features", you will need sample images from the folder images_generate_features_unlabelled, and the models are available in trained_models_identity.**

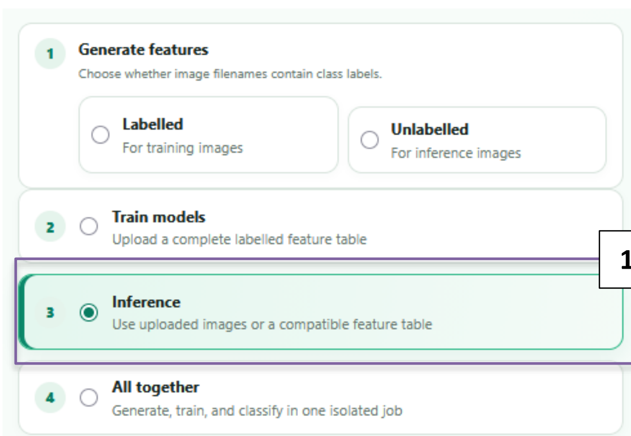
Do step (1) — generate features from the new images:

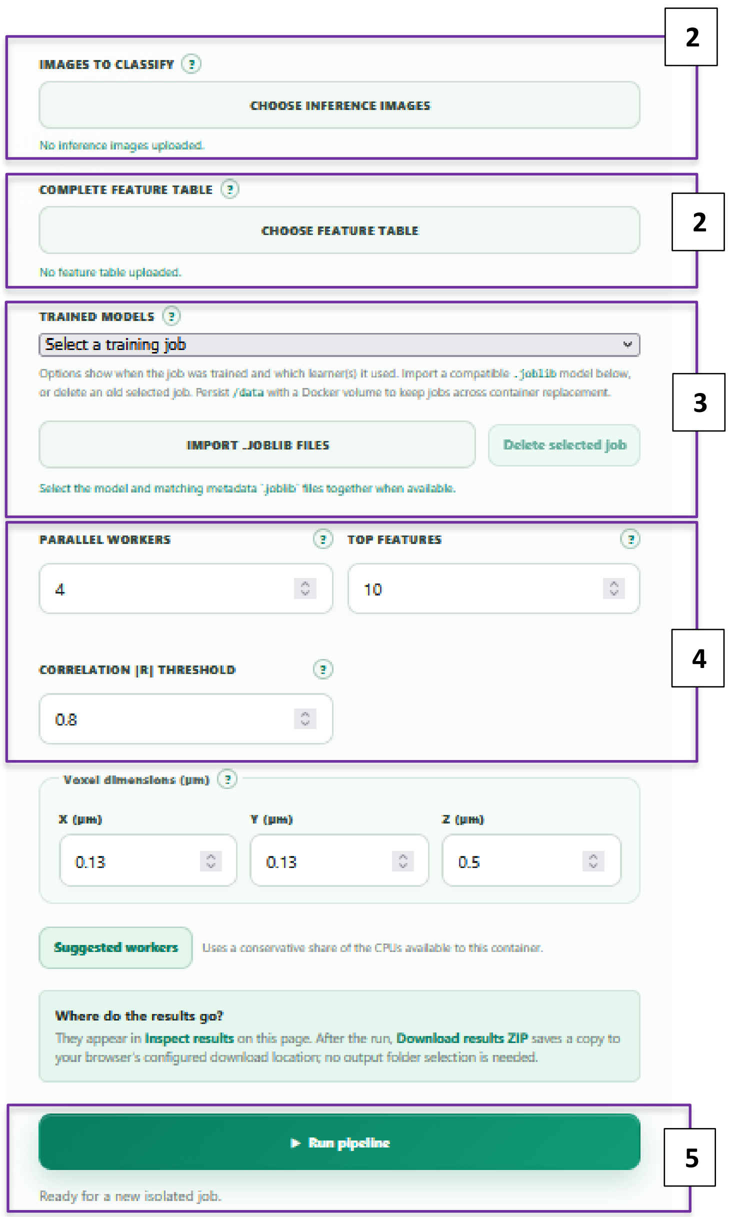
1. Choose "Generate features" in "Unlabelled" mode.
2. Point the software to the folder with the images for inference, using "Choose inference images".
3. Set "Parallel workers" – how much of your CPU to dedicate to the analysis: full capacity when you are not otherwise using the computer, less if you are working on it at the same time. Suggested workers will provide optimal settings for running.
   "Top features" and "Correlation |R| threshold" do not matter here. Adjust the voxel dimensions (the pixel size in micrometres) to match your images.
4. Click "Run pipeline". When the run starts, the software checks the image naming but **does not read labels** – in unlabelled mode the labels are ignored because these images are the unknowns to be classified.



Do step (2) — run the model on those features:

1. Choose “Inference”.
2. Point the software to the folder containing inference images using “Choose inference images” **OR** point the software to the feature table (the features you just generated in unlabelled mode) using “Choose feature table”.
3. Point the software to the folder containing the trained model using the drop-down menu “Trained models”. All trained models are saved in the Docker folder, so use the drop-down menu to choose the appropriate model. Models are saved by the date they were trained, followed by the number of models trained (one when only one model was selected, or six when all learners were selected), and then a code created by Docker for tracking in case of problems. If you would like to delete an old model, or a model you ran as a test, select the model from the drop-down menu and click “Delete selected job”. You can also import trained models using “Import _joblib files”. This takes you to the folder with models; select the models you would like to import (all or just one). Once imported, they will be used for inference.
4. Set “Parallel workers” as before. Suggested workers will provide optimal settings for running.
5. Click “Run pipeline”.





Read the results, in the following order:

1. **inference_summary** — Check the run analysis, e.g. detect incorrectly read labels during training, etc.
2. **rf_predictions** — the predicted class for each image.
3. **rf_probabilities** — the confidence behind each prediction.

Now it’s your turn

- Have fun tonight and let diffuse thinking do the work.
- Tomorrow, read back over what we covered today and highlight anything you did not follow.
- Check the theoretical part to see whether it answers those questions.
- Repeat this exercise.
- Read the paper: https://www.nature.com/articles/s41522-026-01083-8.
- Then move on to the homework, where you run the full dataset and uncover much more detail.